AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 10 · STUDENT LESSON

Gradients with a numerical check

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Before you start

Why is the symmetric-slope denominator twice the step?

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Estimate one partial derivative while other parameters stay fixed. Try the recall before opening notes.

Your learning target

I can estimate one **partial derivative** while other parameters stay fixed.

I can assemble a **gradient** in a declared parameter order.

I can compare numerical and formula based **gradient**s with a **tolerance**.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Slopes and local change. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Parameters and architecture. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Activation functions. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Estimate one **partial derivative** while other parameters stay fixed

Assemble a **gradient** in a declared parameter order

Compare numerical and formula based **gradient**s with a **tolerance**

Retrieve what you already know

> Why is the symmetric-slope denominator twice the step?

My starting explanation

This week's end goal

By the end, I can explain **gradient**s with a numerical check using a traced example and a stated limitation.

Notice the evidence

> Investigate the supplied Gradients with a numerical check case. Separate observations, given facts and assumptions before choosing an explanation.

Use one example with **input** x = 2 and target t = 3. The model predicts y = wx + b, and loss L(w, b) = (2w + b - 3) squared. At w = 1, b = 0, **prediction** is 2, residual is -1 and loss is 1.

There are two parameters in order [w, b].

To estimate the partial slope for w, change w only and keep b = 0. To estimate the partial slope for b, change b only and keep w = 1.

Moving both at once measures change along a combined direction; it does not isolate either **partial derivative**.

Perturbation from [1, 0]	Loss at minus step	Loss at plus step	Symmetric slope
w only with step 0.1	1.44	0.64	-4
b only with step 0.1	1.21	0.81	-2

The **gradient** is [-4, -2] in order [w, b]. Both entries refer to the same original parameter vector, not to a partially updated model.

List the given facts and one fact that is not supplied.

See how it works

> Follow the mechanism for Gradients with a numerical check. Connect each labelled input, deciding step and result.

```
def loss(w, b):
    return (2 * w + b - 3) ** 2

def gradient_check(w, b, step=0.001):
    dw = (loss(w + step, b) - loss(w - step, b)) / (2 * step)
    db = (loss(w, b + step) - loss(w, b - step)) / (2 * step)
    residual = 2 * w + b - 3
    formula = [4 * residual, 2 * residual]
    numerical = [dw, db]
    assert all(abs(a - n) < 1e-6
               for a, n in zip(formula, numerical))
    return numerical

print([round(g, 6) for g in gradient_check(1, 0)])
# [-4.0, -2.0]
```

The helper assumes finite numerical **input**s and a positive nonzero step. The formula follows the squared-loss rule: local loss change is twice the residual times the **prediction** sensitivity.

Changing w by a small amount changes **prediction** twice as much here; changing b changes **prediction** by the same amount.

Exact floating-point equality is not required. Record the step and a **tolerance**, then try more than one parameter setting.

A passing numerical check supports agreement for those tested points; it does not prove all possible **input**s.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

Compare the cases

> Compare the supplied cases for Gradients with a numerical check. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

A w probe has losses 1.44 and 0.64 at 0.9 and 1.1 with b fixed. A learner divides their difference by 0.1. Correct the result.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: Swap the two **gradient** components in the program **output**.

Predicted result

Prediction: The coordinate-wise numerical check exposes the mismatch.

Why the result follows

Reason: The w perturbation changes y twice as strongly as the same b perturbation.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Gradients with a numerical check to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Partial derivative	A local slope with respect to one parameter while others are held fixed
Gradient	The ordered vector of partial derivatives of a scalar function
Finite difference	An estimate of change computed from nearby function values
Tolerance	A stated allowance for numerical disagreement

Use two terms to explain a step in this week’s example.

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Gradients with a numerical check, predict a step, then compare with its explanation.

Two separate probes at one parameter vector

PARAMETER BEING PROBED	HELD FIXED	GRADIENT ENTRY
w near 1	b is 0 and x is 2	First entry -4
b near 0	w is 1 and x is 2	Second entry -2
Assemble in order [w, b]	Same original model	Gradient [-4, -2]

The first two rows are independent local probes, not successive parameter updates. The final row records both results in the declared order.

At w = 0, b = 1, find prediction, residual, loss and the gradient using [4 times residual, 2 times residual].

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Gradients with a numerical check

Check each coordinate of a **gradient** while holding the other parameter fixed.

The facts we will use

Supplied facts

Model $y=2w+b$, target 3. At $[w, b]=[1, 0]$, $y = 2$, residual $y-\text{target}=-1$; loss is residual^2.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/10

Gradients with a numerical check

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Check w direction

Analytic derivative2 x residual x 2=-4.



2. Check b direction

Analytic derivative2 x residual=-2.



3. Numerical check

With step 0.1: w losses 0.64, 1.44 give -4; b losses 0.81, 1.21 give -2.

CHANGE ONE THING

Swap the two gradient components in the program output.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Check w direction	Analytic derivative2×residual×2=-4.
2. Check b direction	Analytic derivative2×residual=-2.
3. Numerical check	With step 0.1: w losses 0.64, 1.44 give -4; b losses 0.81, 1.21 give -2.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A **gradient** lists the slope for each parameter, in a declared order. Changing w by a small amount changes the **prediction** twice as much as changing b by the same amount. That explains the different derivative magnitudes.

Perturb one coordinate while holding the other fixed, then compare the numerical slope with its analytic counterpart.

What this example does not establish

Keep the stated limit

The exact agreement occurs for this quadratic toy case; **finite difference**s in larger models involve rounding and step-size limits.

Keep the stated limit

Read the labelled picture: Why should the numerical checks perturb one coordinate at a time? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Swap the two **gradient** components in the program **output**. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Gradients with a numerical check claim and explain a repair using the supplied evidence.

Check the claim against the facts

ReLU is not differentiable exactly at zero. A symmetric difference across that corner can average two different behaviours.

When checking a network **gradient**, use points whose relevant scores are safely away from zero, and check that the small perturbations do not cross a corner. Document any derivative convention used at zero.

Check the claim against the facts

At the centre of the simple scalar ReLU function, [ReLU(0.1) - ReLU(-0.1)] / 0.2 equals 0.5. This symmetric number does not make the function differentiable at zero.

Also avoid assuming an ever-smaller step is always better: subtraction and rounding can spoil numerical differences.

Locate the earliest unsupported step

Starting at [w, b] = [1, 0], both coordinates move by plus or minus 0.1 together. Compute the symmetric slope and explain why it is not either **partial derivative**.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

Practise Gradients with a numerical check with support, then explain the deciding step in your own words.

For the lesson model at w = 1 and b = 1, calculate **prediction**, residual, loss and **gradient**.

At w = 0, b = 1, find prediction, residual, loss and the gradient using [4 times residual, 2 times residual].

Explain your prediction to a partner before comparing results.

Practise with less help: Catch a denominator error

Use workbook W2 for the connected example "**Gradient**s with a numerical check". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Gradient**s with a numerical check. Proposed explanation: "The **gradient** is one scalar". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why should the numerical checks perturb one coordinate at a time? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Gradients with a numerical check. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

A w probe has losses 1.44 and 0.64 at 0.9 and 1.1 with b fixed. A learner divides their difference by 0.1. Correct the result.

```
def loss(w, b):
    return (2 * w + b - 3) ** 2

def gradient_check(w, b, step=0.001):
    dw = (loss(w + step, b) - loss(w - step, b)) / (2 * step)
    db = (loss(w, b + step) - loss(w, b - step)) / (2 * step)
    residual = 2 * w + b - 3
    formula = [4 * residual, 2 * residual]
    numerical = [dw, db]
    assert all(abs(a - n) < 1e-6
               for a, n in zip(formula, numerical))
    return numerical

print([round(g, 6) for g in gradient_check(1, 0)])
# [-4.0, -2.0]
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	

Record	Your evidence
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Gradients with a numerical check. Explain why each step is justified.

Explain → apply → challenge

For the lesson model at w = 1 and b = 1, calculate **prediction**, residual, loss and **gradient**.

Starting at [w, b] = [1, 0], both coordinates move by plus or minus 0.1 together. Compute the symmetric slope and explain why it is not either **partial derivative**.

Compute the symmetric difference of ReLU at zero with step 0.1. Explain why this does not prove a derivative exists there.

Explain which evidence supports your conclusion and which claim remains unproved.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Image-pattern Network Investigation. Keep an annotated decision and the evidence supporting it.

Add a labelled **gradient**s with a numerical check **evidence** sheet to Image-pattern Network Investigation. Show how you can estimate one **partial derivative** while other parameters stay fixed.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

The **gradient** describes how the stated loss changes near the current parameters. For a differentiable function, moving a sufficiently small distance opposite a nonzero **gradient** decreases the function locally.

The **gradient** alone does not choose a safe finite step, establish a global minimum or demonstrate generalisation.

A numerical checker moves w and b together, gets one slope and copies it into both gradient entries. What is wrong with this check?

Next week combines a **gradient** with an explicit learning rate to update parameters together.

Compute the symmetric difference of ReLU at zero with step 0.1. Explain why this does not prove a derivative exists there.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Gradients with a numerical check case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Two separate probes at one parameter vector

PARAMETER BEING PROBED	HELD FIXED	GRADIENT ENTRY
w near 1	b is 0 and x is 2	First entry -4
b near 0	w is 1 and x is 2	Second entry -2
Assemble in order [w, b]	Same original model	Gradient [-4, -2]

The first two rows are independent local probes, not successive parameter updates. The final row records both results in the declared order.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Check a new example independently

Use x = 3, target 2 and y = wx + b at w = 0, b = 1. With step 0.1, calculate both numerical partials. Compare with [2 times residual times x, 2 times residual].

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Use x = 3, target 2 and y = wx + b at w = 0, b = 1. With step 0.1, calculate both numerical partials. Compare with [2 times residual times x, 2 times residual].

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Gradients with a numerical check, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Check each coordinate of a **gradient** while holding the other parameter fixed.

By the end, I can	How I will show it
Estimate one partial derivative while other parameters stay fixed	Explain the deciding step, show a trace or test, and state one limit.
Assemble a gradient in a declared parameter order	Explain the deciding step, show a trace or test, and state one limit.
Compare numerical and formula based gradients with a tolerance	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Gradients with a numerical check without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Estimate one **partial derivative** while other parameters stay fixed

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building

block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Check each coordinate of a **gradient** while holding the other parameter fixed.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? With step 0.1: w losses 0.64, 1.44 give -4; b losses 0.81, 1.21 give -2.

1. Numerical check

2. Check b direction

3. Check w direction

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Numerical check With step 0.1: w losses 0.64, 1.44 give -4; b losses 0.81, 1.21 give -2. A **gradient** lists the slope for each parameter, in a declared order.

Changing w by a small amount changes the **prediction** twice as much as changing b by the same amount. That explains the different derivative magnitudes.

Perturb one coordinate while holding the other fixed, then compare the numerical slope with its analytic counterpart.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Swap the two **gradient** components in the program **output**. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. The coordinate-wise numerical check exposes the mismatch.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The coordinate-wise numerical check exposes the mismatch. The w perturbation changes y twice as strongly as the same b perturbation.

Repair a missing building block

Review Slopes and local change (ai-5/9). Goal: Calculate an average rate of change over a stated interval. Short practice: Which calculation disproves “any increase helps”?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Parameters and architecture (ai-5/7). Goal: Distinguish architectural choices from parameter values. Short practice: How many new parameters did that one hidden unit add?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Activation functions (ai-5/4). Goal: Trace identity ReLU and sigmoid on a numerical score. Short practice: Which original **input**s cannot be distinguished by this network’s final **output**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

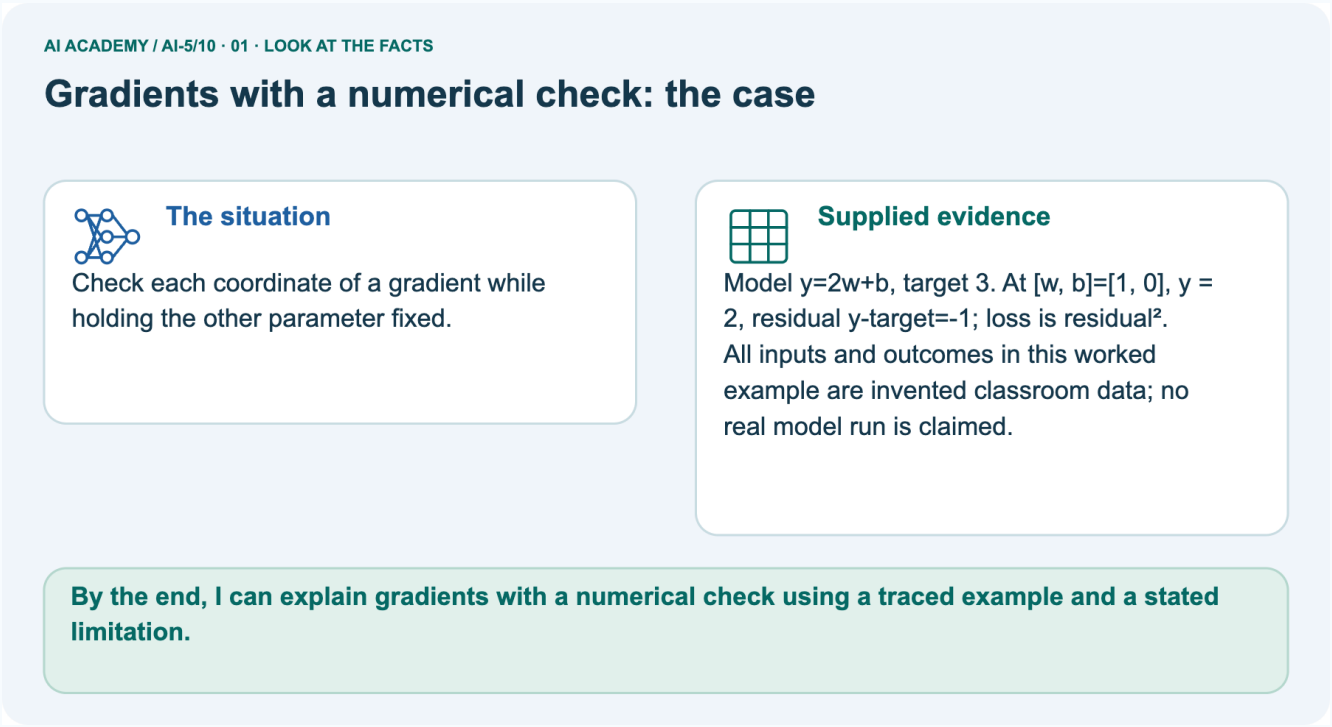
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Gradients with a numerical check graphic using its labels, deciding evidence and relevant limitation.

01. Gradients with a numerical check: the case



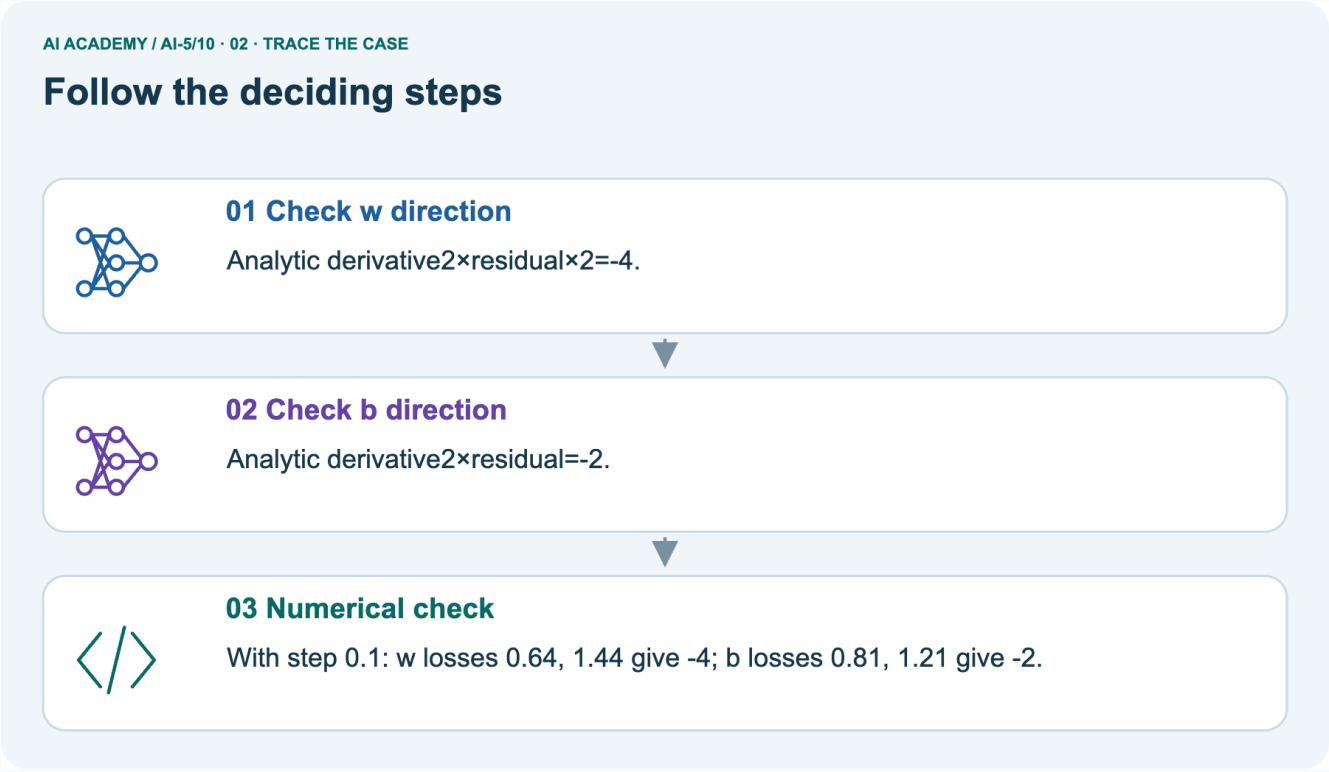
Follow the labelled picture with the matching explanation. Open the vector diagram

Check each coordinate of a **gradient** while holding the other parameter fixed. Supplied facts: Model y=2w+b, target 3. At [w, b]=[1, 0], y = 2, residual y-target=-1; loss is residual².

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Check w direction: Analytic derivative2×residual×2=-4. → Check b direction: Analytic derivative2×residual=-2. → Numerical check: With step 0.1: w losses 0.64, 1.44 give -4; b losses 0.81, 1.21 give -2.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Two separate probes at one parameter vector

AI ACADEMY / AI-5/10 · 03 · INSPECT THE LESSON MODEL

Two separate probes at one parameter vector

Parameter being probed	Held fixed	Gradient entry
w near 1	b is 0 and x is 2	First entry -4
b near 0	w is 1 and x is 2	Second entry -2
Assemble in order [w, b]	Same original model	Gradient [-4, -2]

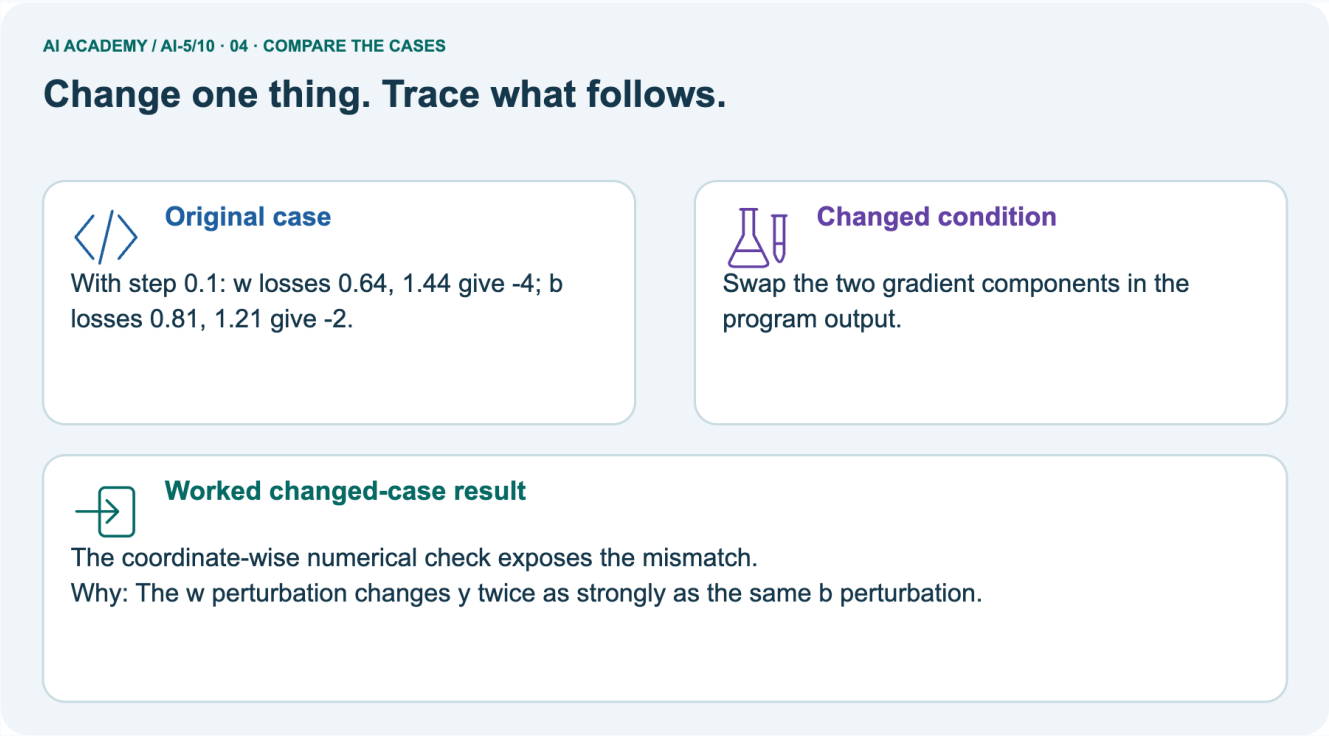
The first two rows are independent local probes, not successive parameter updates. The final row records both results in the declared order.

Follow the labelled picture with the matching explanation. Open the vector diagram

Two separate probes at one parameter vector. w near 1 | b is 0 and x is 2 | First entry -4. b near 0 | w is 1 and x is 2 | Second entry -2. Assemble in order [w, b] | Same original model | **Gradient** [-4, -2]

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



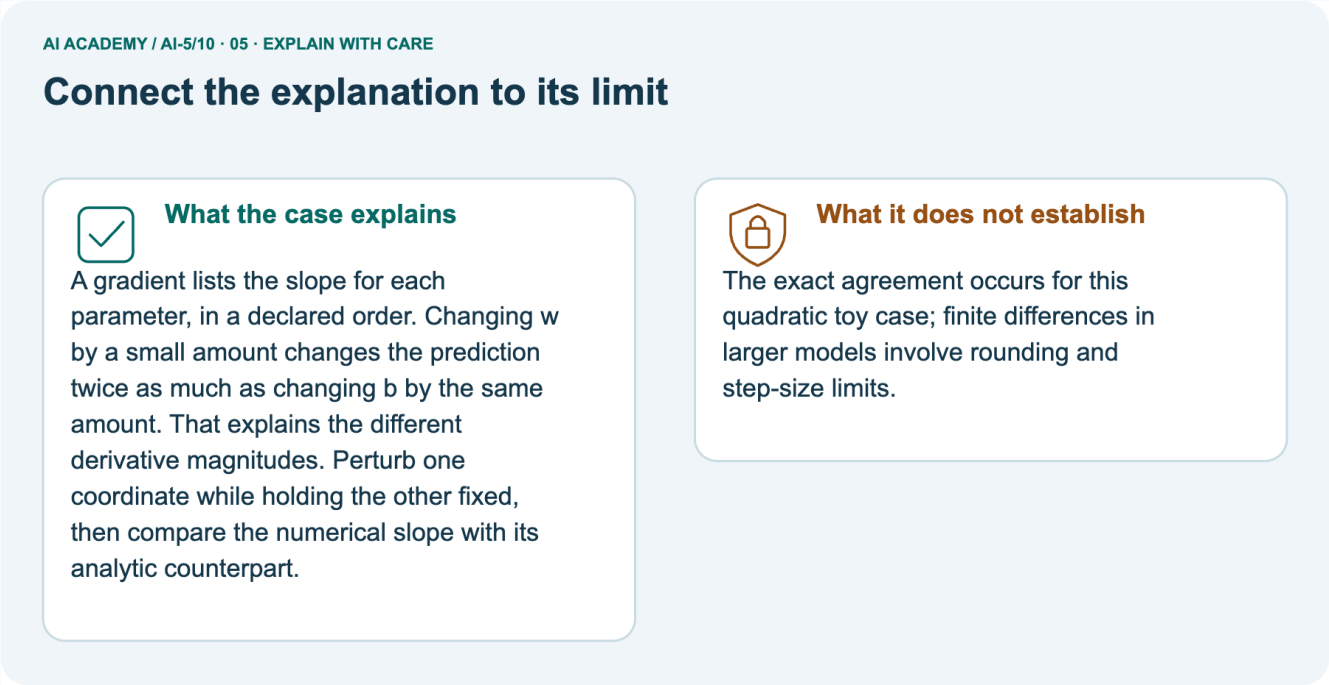
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: With step 0.1: w losses 0.64, 1.44 give -4; b losses 0.81, 1.21 give -2. Changed condition: Swap the two **gradient** components in the program **output**.

Expected result: The coordinate-wise numerical check exposes the mismatch. Reason: The w perturbation changes y twice as strongly as the same b perturbation.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

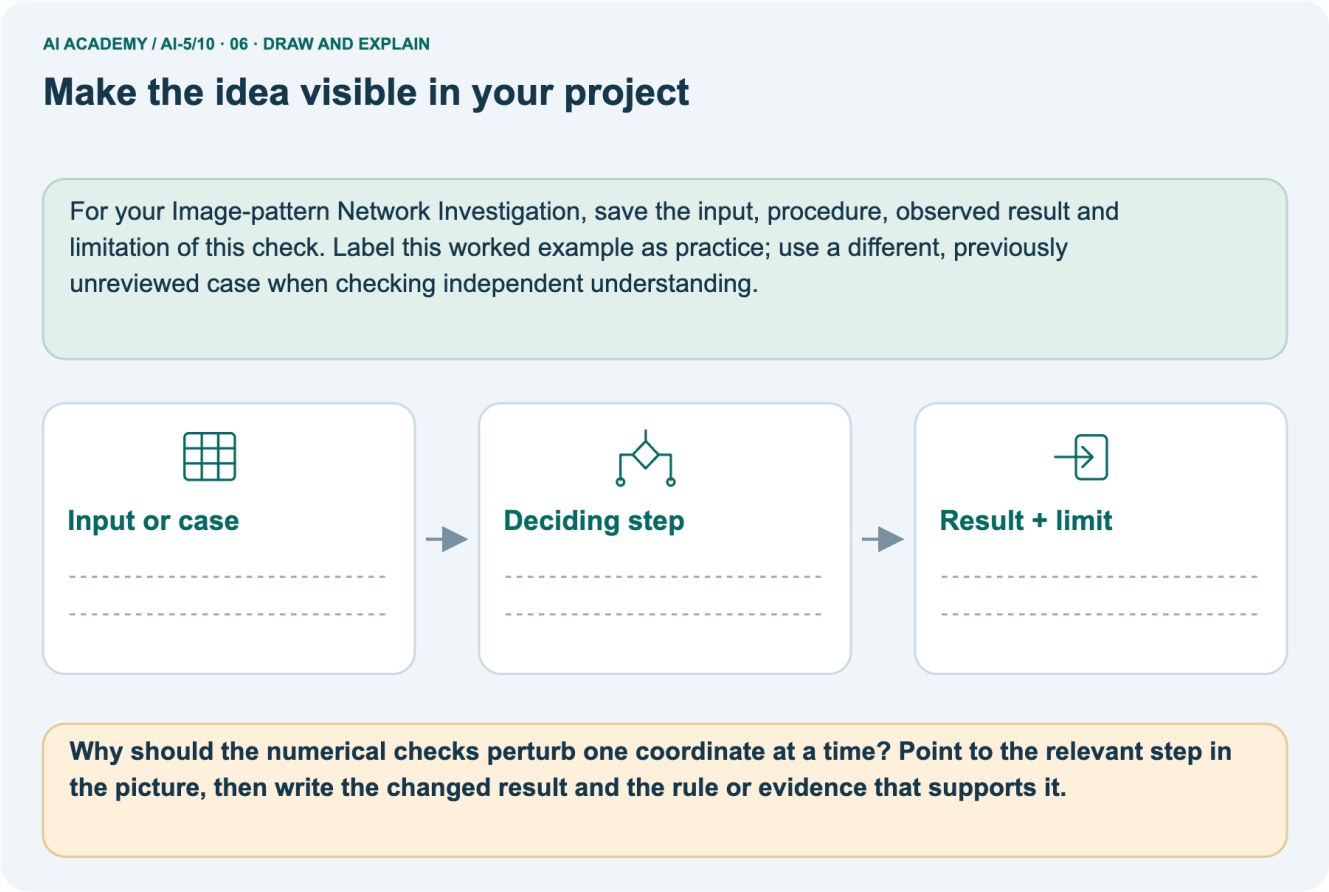
A **gradient** lists the slope for each parameter, in a declared order. Changing w by a small amount changes the **prediction** twice as much as changing b by the same amount. That explains the different derivative magnitudes.

Perturb one coordinate while holding the other fixed, then compare the numerical slope with its analytic counterpart.

Limit: The exact agreement occurs for this quadratic toy case; **finite difference**s in larger models involve rounding and step-size limits.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why should the numerical checks perturb one coordinate at a time? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.